

Problem Set 1

Applied Stats/Quant Methods 1
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Due: October 8, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 8, 2026. No late assignments will be accepted.

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```
1 y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,  
      80, 97, 95, 111, 114, 89, 95, 126, 98)
```

1. Find a 90% confidence interval for the average student IQ in the school (As a hint, you first need the mean and SD to create a CI):

R:

```
1 y_mean <- mean(y)  
2 y_sd <- sd(y)  
3 y_se <- y_sd / sqrt(length(y)) # standard error of the mean  
4 t_value <- qt(0.95, df = length(y) - 1) # confidence coefficient = 0.90  
5 CI_lower <- (y_mean) - t_value * (y_se)  
6 CI_upper <- (y_mean) + t_value * (y_se)
```

Answer: [93.960, 102.92]

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country. Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

$$H_0 : \mu \leq 100$$

$$H_1 : \mu > 100$$

R:

```
1 t_stat = (y_mean - 100) / y_se # t-statistic
2 p_value <- pt(t_stat, df = length(y) - 1, lower.tail = FALSE)
```

Test Statistic (t-value): -0.596

df: 24

p-value: 0.722

Answer:

Since the p-value is greater than 0.05, the null hypothesis cannot be rejected at the 5% significance level. Therefore, there is not enough statistical evidence to conclude that the average student IQ in this school is higher than the national average.

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

State	50 states in US
Y	per capita expenditure on shelters/housing assistance in state
X1	per capita personal income in state
X2	Number of residents per 100,000 that are "financially insecure" in state
X3	Number of people per thousand residing in urban areas in state
Region	1=Northeast, 2= North Central, 3= South, 4=West

Explore the `expenditure` data set and import data into R.

```
1 CI_upper <- (y_mean) + t_value * (y_se)
```

- Please plot the relationships among Y , $X1$, $X2$, and $X3$? What are the correlations among them (you just need to describe the graph and the relationships among them)?

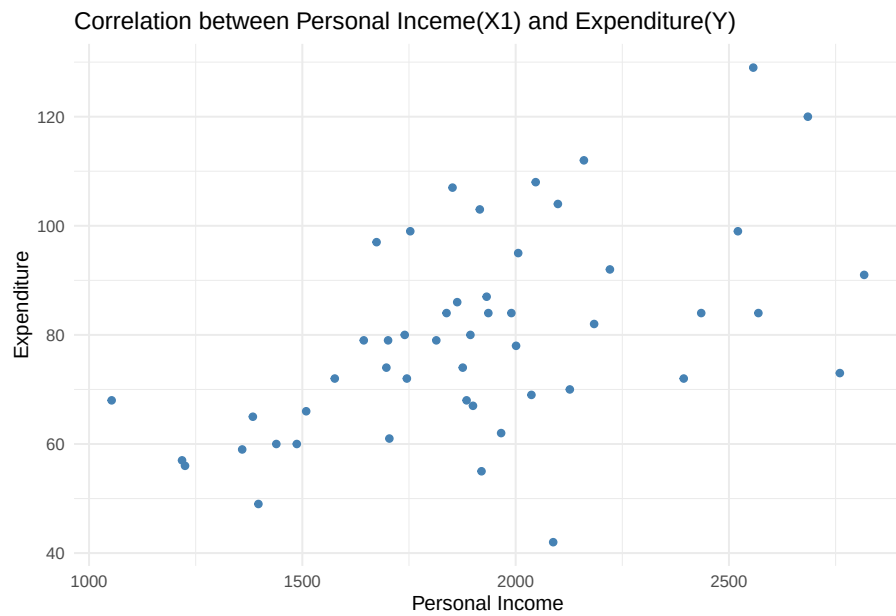
R:

```
1 #Question 1
2 p1 <- ggplot(expenditure, aes(x = X1, y = Y)) +
3   geom_point(color = "steelblue") +
4   labs(x = "Personal Income", y = "Expenditure", title = "Correlation
5     between Personal Income(X1) and Expenditure(Y)") +
6   theme_minimal()
7 p2 <- ggplot(expenditure, aes(x = X2, y = Y)) +
8   geom_point(color = "darkgreen") +
9   labs(x = "Financially Insecure", y = "Expenditure", title = "
10     Correlation between Financial Insecurity(X2) and Expenditure(Y)") +
11   theme_minimal()
12 p3 <- ggplot(expenditure, aes(x = X3, y = Y)) +
13   geom_point(color = "coral") +
14   labs(x = "Urban Population", y = "Expenditure", title = "Correlation
15     between Urban Population(X3) and Expenditure(Y)") +
16   theme_minimal()
17 expenditure_cor <- cor(expenditure[,c("Y", "X1", "X2", "X3")])
```

Result:
Correlation Matrix

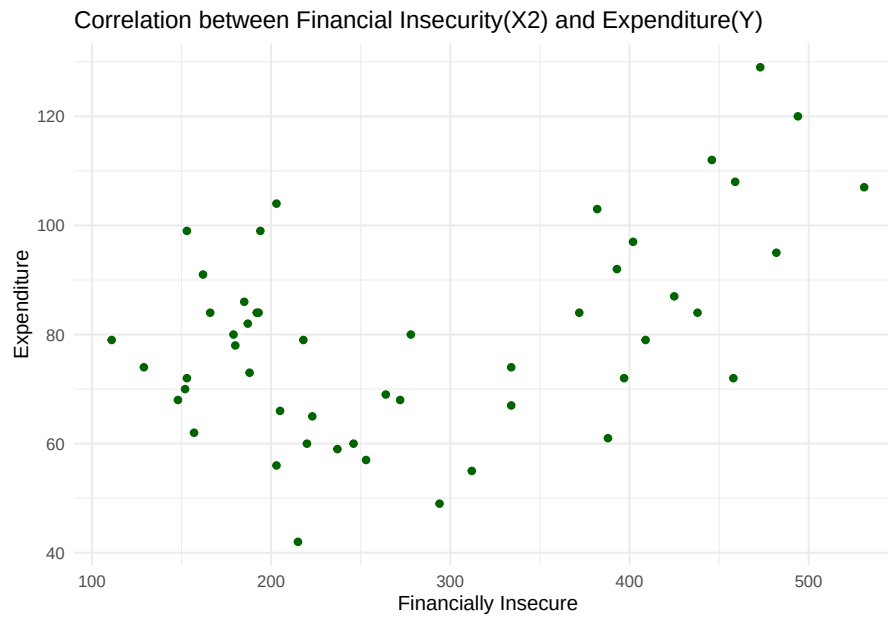
	Y	X1	X2	X3
Y	1.0000000	0.5317212	0.4482876	0.4636787
X1	0.5317212	1.0000000	0.2056101	0.5952504
X2	0.4482876	0.2056101	1.0000000	0.2210149
X3	0.4636787	0.5952504	0.2210149	1.0000000

Figure 1



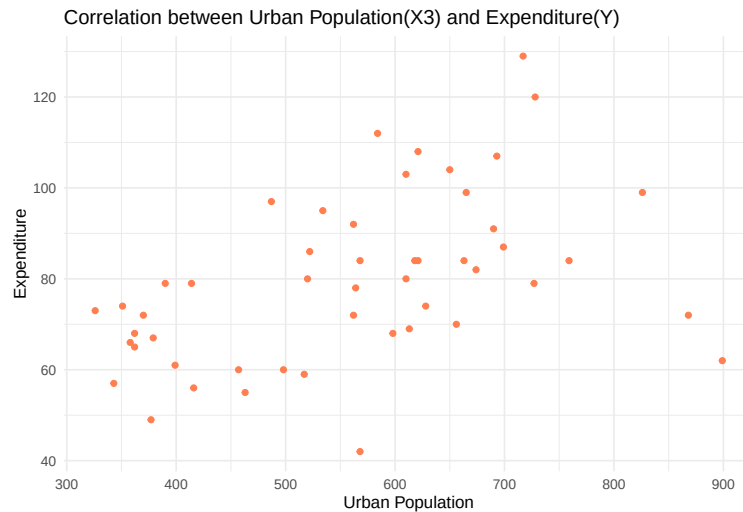
As the per capita expenditure on shelters/housing assistance (Y) increases, per capita personal income (X1) also tends to increase. The correlation coefficient (0.531) indicates a moderate positive correlation between Y and X1.

Figure 2



When the number of financially insecure residents per 100,000 (X2) is below about 300, per capita expenditure on shelters/housing assistance (Y) tends to decrease as X2 increases. However, when X2 is above 300, Y tends to increase as X2 increases. The correlation coefficient (0.448) indicates a moderate positive correlation between Y and X2 overall.

Figure 3



As the per capita expenditure on shelters/housing assistance (Y) increases, number of residents per thousand residing in urban areas (X3) also tends to increase. The correlation coefficient (0.448) indicates a moderate positive correlation between Y and X3.

- Please plot the relationship between Y and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

R:

```

1 expenditure$Region <- factor(
2   expenditure$Region,
3   levels = c(1, 2, 3, 4),
4   labels = c("Northeast", "North Central", "South", "West")
5 )
6
7 boxplot_expenditure <- ggplot(expenditure, aes(x = factor(Region), y = Y)) +
8   geom_boxplot() +
9   labs(title = "Expenditure on Shelters/Housing Assistance by Region", x
10    = NULL, y = "Expenditure")
11 boxplot_expenditure
12
13 #Question 3
14 p4 <- ggplot(expenditure, aes(x = X1, y = Y)) +
15   geom_point(aes(color = factor(Region), shape = factor(Region))) +
16   labs(x = "Personal Income", y = "Expenditure", title = "Correlation
17    between Personal Income(X1) and Expenditure(Y) by Region") +

```

Figure 4

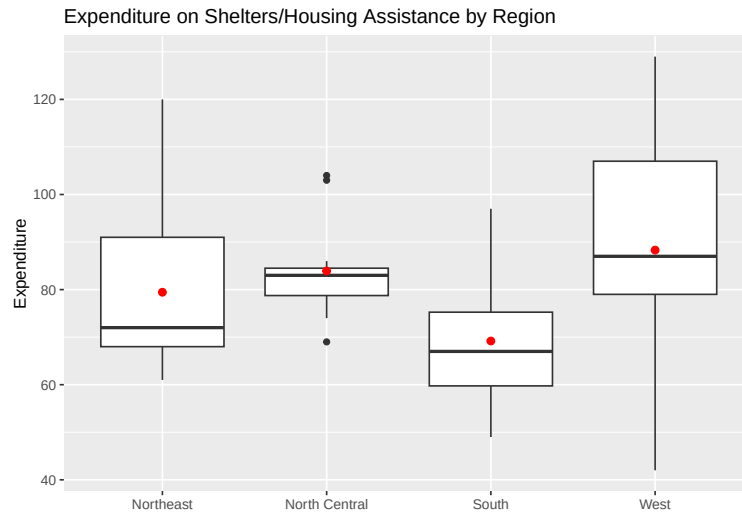


Figure 4 shows the distribution of per capita expenditure on shelters/housing assistance (Y) across different regions. The red dots in the boxplots represent the average expenditure for each region. The West region has the highest average per capita expenditure on shelters/housing assistance.

- Please plot the relationship between Y and $X1$? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

R:

```
1 p4 <- ggplot(expenditure, aes(x = X1, y = Y)) +
2   geom_point(aes(color = factor(Region), shape = factor(Region))) +
3   labs(x = "Personal Income", y = "Expenditure", title = "Correlation
4     between Personal Income(X1) and Expenditure(Y) by Region") +
5   theme_minimal()
```

Figure 5



As Figure 1 suggests, there is a positive correlation between per capita expenditure (Y) and personal income (X1). When Region is added as another variable, Figure 5 shows that the relationship between X1 and Y seems less clear in the North Central and West regions compared with the Northeast and South regions.